

Oriented nanoporous MOFs to mitigate polysulfides migration in lithium-sulfur batteries

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ABSTRACT

Metal-organic frameworks (MOFs) have several attractive features for energy applications including tunable pore sizes, highly-ordered structures and versatile chemical reactivity. Here, we show the antiferroelectric perovskite dimethylammonium zinc formate (DMAZF) MOF $[(\text{CH}_3)_2\text{NH}_2] \text{Zn} (\text{HCO}_2)_3$ as an effective molecular sieve to mitigate polysulfide (PS) migration in lithium-sulfur batteries (LSBs) when combined with conductive carbon nanotubes (CNTs). The DMAZF was selected due to both its nanopore structure and the presence of the Zn-metal site. The nanopores facilitate PS physical separation, whereas the Zn-site acts as a Lewis-acid site to attract the PS and also as a catalytic site to encourage electrochemical redox reactions. The hybrid DMAZF/CNTs/sulfur electrode, with 5 mg cm^{-2} sulfur loading, delivers an initial high specific capacity of 1260 mAh g^{-1} at 0.05C and 1007 mAh g^{-1} at 0.1C with the degradation of only 0.07% after 120 cycles. Even at 7 mg cm^{-2} sulfur loading, the electrode performance decreases only 0.12% per cycle even after 500 cycles at 0.5C.

1. Introduction

Lithium-sulfur batteries (LSBs) promise both a high theoretical specific capacity of sulfur (1675 mAh g^{-1}) and gravimetric energy density (2600 Wh kg^{-1}) as compared to other energy storage alternatives [1–3]. Apart from this, the environmental friendliness, low cost and natural abundance of sulfur is pushing LSBs research into the limelight [4,5]. The development of LSBs, has a number of intrinsic challenges including the insulating nature of sulfur ($5 \times 10^{-30} \text{ S cm}^{-1}$ at 25°C) and its volume expansion during cycling [6]. More prominently, the well-known shuttle effect, caused by the migration of soluble polysulfides (PS) from the cathode to the anode during charging/discharging, leads to fast capacity loss. In addition, sluggish sulfur redox reactions further render large overpotential losses [3,7] which limits the LSBs performance. The aforementioned limitations of LSBs simultaneously lead to limited sulfur utilization, rapid capacity fading leading to short life cycles [8,9]. The development of LSBs technology requires suitable high-performance electrode materials.

Recently, the electronically conductive porous materials and the

physical confinement of sulfur within their scaffold have been widely employed for high-performance LSBs [10,11]. However, the ability of physical adsorption of PS inside non-polar porous carbon due to the van der Waals interaction is often weak (0.1–0.7 eV) [12–14]. Materials that provide strong and abundant binding sites in combination with good electron and ion pathways are demonstrated to be more suitable as cathode hosts for high capacity and stable LSBs [15–25]. The catalytic activity of these functionalized host materials significantly promotes the PS conversion in LSBs [26–29]. MOFs represent a class of porous materials, composed of metal ions and organic ligands are also suitable candidates to meet the needs of next-generation energy storage technologies [30–34]. MOFs have also been used for LSB research due to their high porosity, large specific surface area, tunable porosity and versatile functionality affording both catalytic and chemical interactions to retard PS migration [35–37]. Zhou et al. [38] have shown the significance of conductive MOF channels to confirm the access of Li^+ and electron to sulfur for successive reactivation during cycling. Li et al. [39] introduced a systematic comparison of the different pore sizes of Y-FTZB, ZIF-7, ZIF-8, and HKUST-1 to mitigate the PS diffusion and

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achieved a stable LSB performance. Bai et al. [40] demonstrated MOF-based separator as an effective ionic sieve to efficiently suppress the undesired PS migrating to the anode side. Xu et al. [41] have shown how MOFs need to be designed to facilitate Lewis acid-base interaction with the PS. They propose that Lewis-acid metallic centers such as Zn, Mn, Fe, Co, Ni and Cu are inclined to coordinate with soluble PS anions as axial ligands and can effectively trap the soluble PS within the cathode [30,36,42]. Basically, the Lewis-acid metal center is inclined to coordinate with soluble PS anion through Lewis-acid interactions. Zheng et al. [30] has established a very good example of Lewis-acid interactions between MOFs and PS. MOFs can provide a wide range of opportunities to alleviate the PS shuttle through both physical and chemical interactions to immobilize the PS. Furthermore MOFs can be designed to enhance electrochemical kinetics and also provide uniform nucleation sites for the solid discharge products [27]. Want et al. has demonstrated that the single atomic Co significantly accelerate the Li^+ diffusion in LSBs [43]. Therefore, the ideal PS host needs to have highly conductive metals with suitable pores for fast Li^+ transportation, and adsorptive capacity of PS [44].

Herein, we report the use of the antiferroelectric perovskite DMAZF MOF with a Zn-metal site and tuned porosity as an effective sulfur host for high-performance LSBs. This carefully selected network has many advantages to enable high-performance LSBs. Firstly, the nanoporous structure can act as a physical trap and a molecular sieve facilitating Li^+ migration while blocking PS migration (Fig. 1d) [45,46]. Secondly, the electrolyte can be soaked into the MOF pore structure, which can maintain good Li^+ conduction at a low E/S. Thirdly, the Zn-metal site acts as a Lewis-acid site to attract the PS and also to encourage electrochemical kinetics of the nucleated PS [27,47]. The newly developed DMAZF MOF exhibited excellent electrocatalytic activities for LSBs supported by conductive CNTs. The DMAZF/CNTs/sulfur cathode shows a high specific capacity of 1260 mAh g^{-1} at 0.05C and 1007 mAh g^{-1} at 0.1C with a retention capacity of 913 mAh g^{-1} after 120 cycles at 5 mg cm^{-2} sulfur loading with a fading of 0.07% per cycle. Even at 7 mg cm^{-2} sulfur loading, the cathode initiates a fading of 0.12% per cycle even after 500 cycles at 0.5C rate.

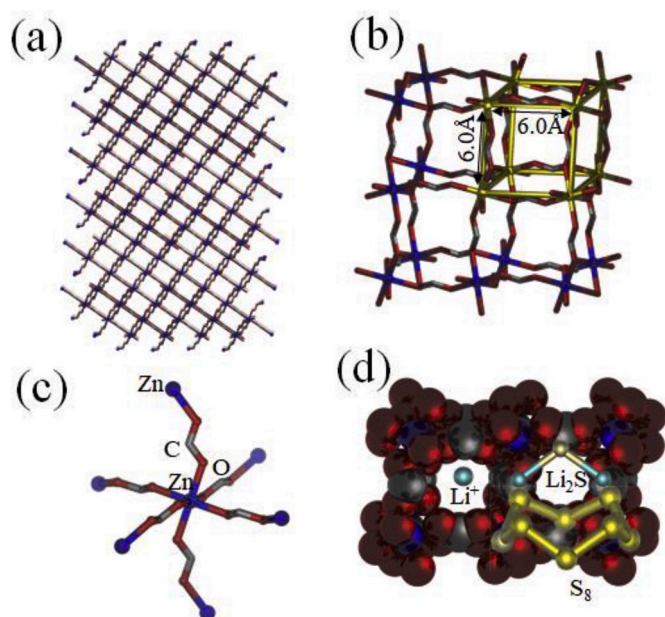


Fig. 1. The representation of the crystal structure of the network DMAZF where (a) crystal structure packing. (b) The network window size to access the void space. (c) metal node showing the coordination of carbon and oxygen with Zn. (d) Illustrates the mechanism of PS mitigation process and access of Li^+ through nano-pores.

2. Experimental

2.1. Materials preparation

2.1.1. Preparation of DMAZF

Formic acid (2.0 g, 44 mmol) was added to a solution of zinc nitrate hexahydrate (6.6 g, 22 mmol) in 50 mL dimethylformamide (DMF). The mixture placed in a Teflon sealed vessel and was heated slowly to 120°C at a rate of 5°C per hour, the temperature was held at 120°C for 72 h before was allowed to reach room temperature during 24 h to produce colourless crystals suitable (4.4 g, yield 64%). The solid was then washed several times with DMF, water, methanol and then with acetone followed by drying the product under high vacuum for 24 h. Elemental analysis calculated for $\text{Zn}_2(\text{HCOO})_4$: C = 15.66, H = 1.3%; found C = 15.34, H = 1.32%.

2.1.2. Pyrolysis of UIO-66 and Br-UIO-66

The UIO-66 [48] and the modified Br-UIO-66 [49] were prepared by following previously reported procedures and the PXRD data match the reported data. The prepared networks were washed several times with DMF, water, methanol and then with acetone followed by drying the product under high vacuum for 24 h before they were used.

2.1.3. Preparation of sulfur electrode and electrolyte

To prepare the cathode, the DMAZF, UIO-66, and Br-UIO-66 were composited with the CNTs at a ratio of 3:2. Subsequently, each of these materials was mixed with 70 wt % sulfur to prepare DMAZF/CNTs/Sulfur, UIO-66/CNTs/Sulfur and Br-UIO-66/CNTs/sulfur through the infiltration process at 155°C . The UIO-66 and Br-UIO-66 without and with Br metal respectively have been selected with a larger pore diameter to compare with the DMAZF.

The 80 wt % of each of these composited were mixed, respectively, with 10 wt % of conductive Super P and 10 wt % of PVDF for electrodes preparation followed by slurry coating doctor blade process on the carbon coated aluminum foil. The sulfur cathode was cut into a circular pellet with a diameter of 10 mm.

The electrolyte solution was prepared using a 1, 3 dioxolane (DOL) and 1, 2 dimethoxymethane (DME) at the volumetric ratio of 1:1, 1 M lithium bis (trifluoromethane sulfonyl) imide (LiTFSI) and 0.1 M LiNO_3 . The electrolyte was prepared inside a glovebox, supplied with Ar, with O_2 and H_2O levels below 0.1 ppm to avoid air and moisture contamination.

2.1.4. Battery assembly

The LSBs were fabricated inside an Ar-filled glove box with a maximum of 0.1 ppm of H_2O and O_2 . A lean electrolyte/sulfur ratio of $6 \mu\text{L/mg}$ was used to assemble the LSBs. A hydraulic crimping machine (MSK-110 Crimper) was used to seal the LSBs for electrochemical testing.

2.1.5. Electrochemical measurements

The LAND system was used to acquire the electrochemical performance of LSBs within a voltage window of $1.7\text{--}2.8 \text{ V}$ (Li/Li^+). The Biologic VSP also was used to capture the CV profiles of the inactivated and activated cells at 0.15, 0.20, 0.25, 0.3 mV s^{-1} , respectively.

2.1.6. Characterization tools

The morphology and structure of the inactivated and activated samples were characterized using a field emission transmission electron microscopy (FE-TEM F20) and a scanning electron microscope (SEM) (SU-3500) is equipped with an Oxford X-MAXN energy-dispersive X-ray spectroscopy (EDS) detector. XPS survey data was acquired using a Kratos Axis ULTRA X-ray photoelectron spectrometer incorporating a 165 mm hemispherical electron energy analyzer. The incident radiation was monochromatic $\text{Al K}\alpha$ X-rays (1486.6 eV) at 150 W (15 kV, 10 mA). Survey (wide) scans were taken at an analyzer pass energy of 160 eV and

multiplex (narrow) high-resolution scans at 20 eV. Survey scans were carried out over 1200–0 eV binding energy range with 1.0 eV steps and a dwell time of 100 ms. Narrow high-resolution scans were run with 0.05 eV steps and at a dwell time of 250 ms. Base pressure in the analysis chamber was 1.0×10^{-9} torr and during sample analysis 1.0×10^{-8} torr. Atomic concentrations were calculated using the CasaXPS version 2.3.14 software and a Shirley baseline with Kratos library relative sensitivity factors (RSFs). Peak fitting of the high-resolution data was also carried out using the CasaXPS software. N_2 adsorption-desorption isotherms were measured using a Tristar II (Micromeritics Vac prep-061). The surface area and pore size distribution of the samples were measured by the Brunauer-Emmett-Teller (BET) method and non-local density functional theory (NLDFIT) methods, respectively. Thermal gravimetric analysis (PerkinElmer, Diamond TG) (TGA) for CNT-70 was performed in a nitrogen atmosphere to measure wt % mass loss at a heating rate of $10^\circ\text{C min}^{-1}$ till 800°C . The XRD (X-Ray Diffraction) of samples were analyzed on a Bruker D8 powder X-Ray diffractometer equipped with a Cu source, operated at 40 kV and 40 mA, and energy discriminating 2D array detector that minimizes a fluorescent background. Data were collected using a Bragg-Brentano geometry with a 0.12° -degree divergence slit from 5° to 80° two-theta at a resolution of 0.02° and at 1.2 s per step.

3. Results and discussion

DMAZF is formed by the reaction of formic acid with zinc nitrate hexahydrate. The single-crystal 3D network DMAZF is shown in Fig. 1a&c. The octahedral Zn^{II} coordinates to six bridging formate ligands to form a cube-like 3D framework. DMAZF encapsulates a void space of 21% of the unit cell ($545.26 \text{ \AA}^3$) occupied by dimethylammonium (DMA) cations which is accessible via a window size of approximately $6 \text{ \AA}$ (Fig. 1b). This window size is even less than the reported HKUST-1 MOF by Bai et al. [40] This type of network can as such act as a molecular sieve - blocking PS migration while permitting the Li^+ diffusion (Fig. 1d).

The morphology of DMAZF is characterized by scanning electron microscopy (SEM) and transmission electron microscopy (TEM) as shown in Fig. 2. The SEM morphology shows a layered structure with pores sizes ranging from around 100 nm to 1 μm in size (Fig. 2a). The

surface morphology of UIO-66 and Br-UIO-66 is compared in Fig. S1. The TEM image of the DMAZF shows the combination of both large and small scale nanopores (Fig. 2b). These nanopores are randomly oriented and interconnected. After the heat treatment of DMAZF with the CNTs/sulfur at 150°C , the sulfur is melted and impregnated inside the nanopores, which provides an even sulfur coverage throughout the DMAZF (Fig. 2c). The STEM Energy-dispersive X-ray spectroscopy (EDS) mapping of the DMAZF/Sulfur/CNTs are introduced in Fig. 2d–h, where, the distribution of carbon, Oxygen, Zinc, and sulfur can be visualized. The experimental powder X-ray diffraction (PXRD) of the raw DMAZF with the DMAZF/CNTs/Sulfur is shown in Fig. 2i. The most prominent peak for the raw DMAZF is corresponding to the (112) plane. The intensity of the 112 crystal peak is further reduced in the PXRD of DMAZF/CNTs/Sulfur, which is consistent with the TEM image shown in Fig. 2b. Brunauer-Emmett-Teller (BET) tests show the surface areas of Br-UIO-66 and DMAZF are $230.10 \text{ m}^2 \text{ g}^{-1}$ and $29.15 \text{ m}^2 \text{ g}^{-1}$, respectively (Fig. S2). The DMAZF shows pore distribution below 5 nm, whereas the Br-UIO-66 shows larger pores.

To trace the elemental composition of DMAZF and Br-UIO-66, the XPS analysis was performed and shown in Fig. 3a. The presence of Zn 2p, O 1s and C 1s are found in the DMAZF. On the other hand, the Br-UIO-66 shows the presence of Zr 3d, Br 3d, O 1s and C 1s. It is noteworthy to highlight here that metals such as Zn, Zr and Br acts as Lewis-acid sites and electrocatalysts to accelerate the PS redox reaction [30]. Moreover, implanting of these metals within the skeleton of mesoporous carbon simultaneously improves the interaction with PS and maintains the mesoporous structure [50]. In the high-resolution XPS spectrum of Zn 2p, Zr 3d and Br 3d are deconvoluted and shown in Fig. 3b–d. The high-resolution XPS Zn 2p spectra confirm the presence of prominent peaks centered at a binding energy of $\sim 1021.0 \text{ eV}$ - indicating Zn 2p_{3/2} (Fig. 3b) [51]. Four peaks are used to fit the high-resolution spectra of Zr 3d and Br 3d shown in Fig. 3c–d. As can be seen from Zr 3d, the peaks at $\sim 181.0 \text{ eV}$, $\sim 182.5 \text{ eV}$, $\sim 183.6 \text{ eV}$ and $\sim 184.9 \text{ eV}$ can be attributed to the Zr 3d_{5/2}, Zr 3d_{5/2}, Zr 3d_{3/2} and Zr 3d_{3/2}, respectively [52]. On the other hand, the peaks at $\sim 68.3 \text{ eV}$, $\sim 70.5 \text{ eV}$, $\sim 69.6 \text{ eV}$ and $\sim 71.5 \text{ eV}$ can be attributed to the Br 3d_{5/2}, Br 3d_{5/2}, Br 3d_{3/2} and Br 3d_{3/2}, respectively [53]. It is worth to highlight that these metals are beneficial to capturing PS and enhancing the rate of chemical conversion due to their catalytic activity [27].

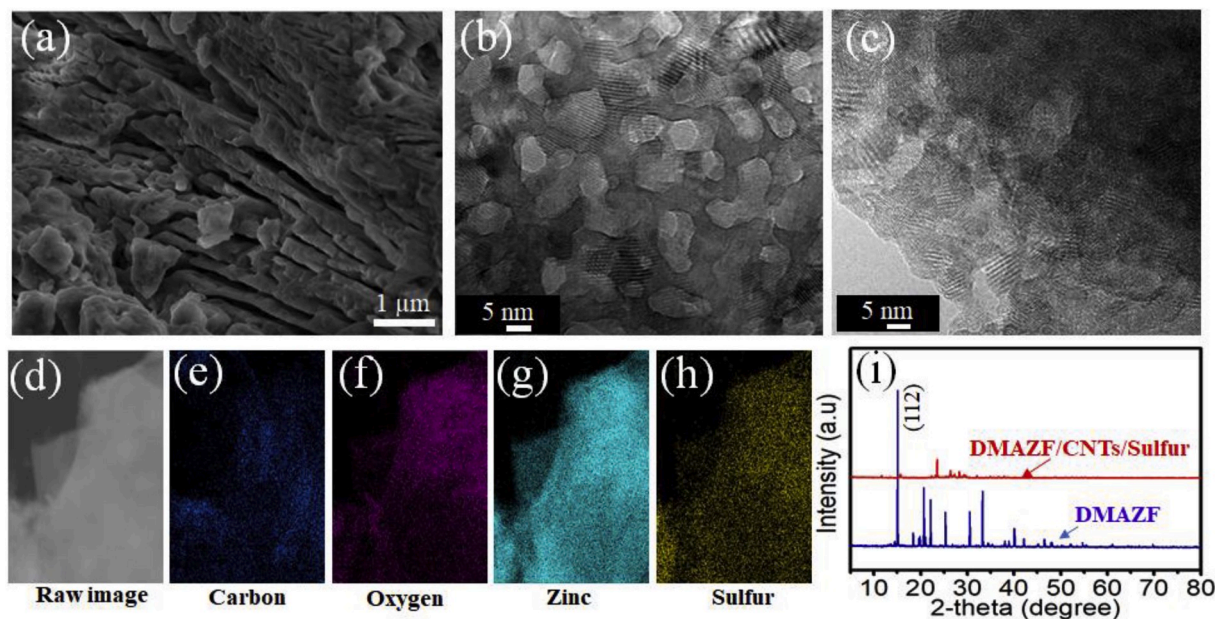


Fig. 2. (a) SEM images of DMAZF, shows a bundle of layers with porous structure. (b) HRTEM image of DMAZF samples shows an interconnected nanoporous structure. (c) HRTEM image of DMAZF/CNTs/sulfur composite. (d–h) raw STEM image with the mapping for carbon, oxygen, zinc, and sulfur. (i) Powder x-ray diffraction pattern for DMAZF raw materials in comparison with the DMAZF/CNTs/sulfur composite.

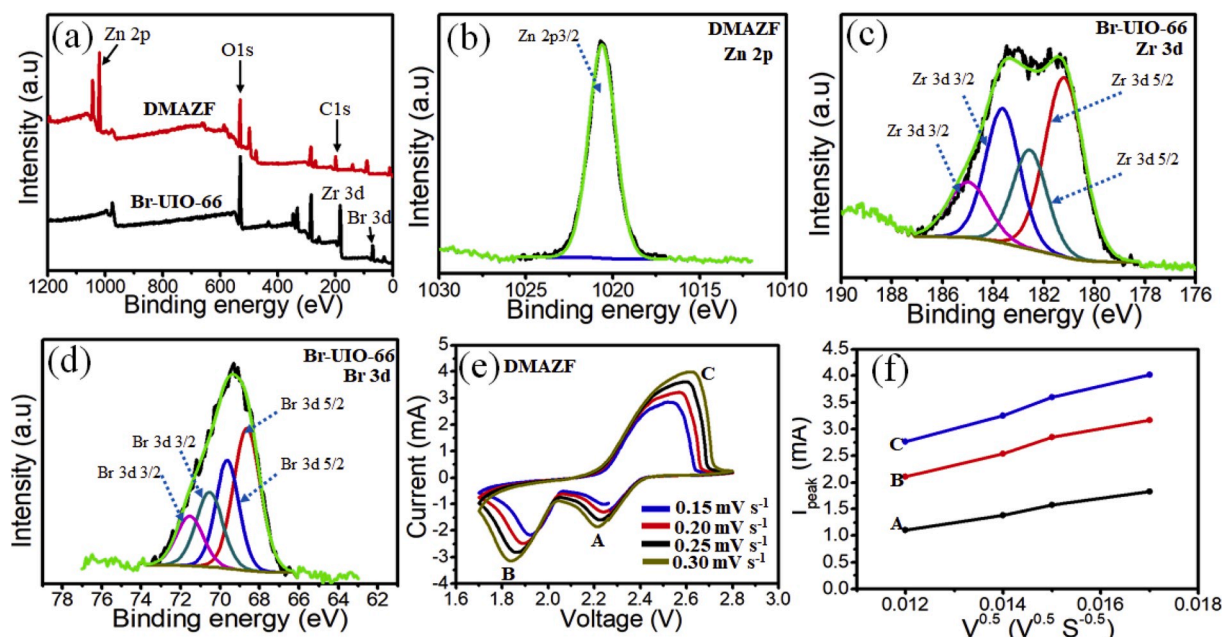


Fig. 3. (a) XPS survey of DMAZF (red) and Br-UIO-66 (black). (b) high-resolution scan of Zn 2p of DMAZF and (c–d) Br 2p and Zr 2p, respectively, from Br-UIO-66. (e) Cyclic voltammograms at different C-rate for DMAZF/CNTs/Sulfur electrode-based LSBs and its rate-dependent diffusion curves shown in (f). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Fig. 3e shows the CV curves of DMAZF/CNTs/Sulfur electrode contained LSBs at different scans of 0.15, 0.2, 0.25 and 0.3 mV s^{-1} , respectively. In all these scans, two-step redox peaks (A and B) are observed during the lithiation/delithiation processes of sulfur and a continuous oxidation peak (C). The two reduction peaks (A and B) correspond to the two discharge plateaus, which result from the transformation processes from S to Li_2S_x ($x = 4-8$) and Li_2S_4 to $\text{Li}_2\text{S}_2/\text{Li}_2\text{S}$, respectively. The oxidation peak (C) implies to the transformation of $\text{Li}_2\text{S}_2/\text{Li}_2\text{S}$ to $\text{Li}_2\text{S}_8/\text{S}$ [54]. Further, the Li-ion diffusion coefficient (D_{Li}^+) were calculated based on the Randles–Sevcik equation in conjunction with the CV curves at different scan rates. The D_{Li}^+ values of A, B, and C peaks are, 0.5, 1.07 and $1.49 \times 10^{-7} \text{ cm}^2 \text{ s}^{-1}$, respectively. The experimental D_{Li}^+ values are comparatively larger than the reference values reported in the literature [54–56]. This indicates that the DMAZF/CNTs/sulfur electrode facilitates ion diffusion, which facilitates subsequent reaction kinetics. This enhances the LSB electrochemical performance [54,55,57].

The DMAZF is introduced for the first time in LSBs systems. Hence, authors have selected two different MOFs as UIO-66 and Br-UIO-66 to compare with the DMAZF with the similar experimental conditions. The electrochemical performance of the DMAZF as a molecular sieve was compared to UIO-66 [48] and Br-UIO-66 [49]. The DMAZF has micropores (Fig. S2c) whereas Br-UIO-66 has mesopores (Fig. S2d). The aim is to understand the impact of the DMAZF with micropores and Zn-metal over the Br-UIO-66 which has Br and mesopores to physically and chemically confine the PS in LSBs. The superiority of the DMAZF in comparison to the existing MOFs with their electrochemical performances could be found in Table S1 and in Ref. [11].

Before testing the battery cycling performances, the active sulfur was impregnated into the porous medium of 3:2 ratios hybrids of DMAZF/CNTs, Br-UIO-66/CNTs and UIO-66/CNTs by a typical melt-diffusion process [58] at 155 °C to obtain the DMAZF/Sulfur/CNTs, Br-UIO-66/Sulfur/CNTs and UIO-66/Sulfur/CNTs, respectively. The thermogravimetric analysis (TGA) curves of DMAZF/CNTs/Sulfur and Br-UIO-66/CNTs/Sulfur samples are shown in the supporting information with their respective SEM images (Figs. S3 and S4). It can be noted that both raw DMAZF and Br-UIO-66 losses some weight around at 155 °C, however, in the presences of CNTs/Sulfur the structure become

stable with no weight loss found at 155 °C. The details of the electrode preparation process are given in the experimental section (Supporting information).

The corresponding electrochemical performances of these three different host materials are shown in Fig. 4a. Here the measurements show that both the DMAZF/Sulfur/CNTs cathodes can provide much higher electrochemical performance as compared to Br-UIO-66 and UIO-66 assisted sulfur cathodes. The DMAZF/Sulfur/CNTs delivered a remarkable capacity of 1260 mAh g^{-1} at 0.05C and 5 mg cm^{-2} sulfur loading. At the similar sulfur loading, it initiates 1007 mAh g^{-1} with a fading of 0.07% per cycle after 120 cycles at 0.1C. On the other hand, at 0.1C, the Br-UIO-66/Sulfur/CNTs and UIO-66/Sulfur/CNTs electrodes exhibits 927 mAh g^{-1} and 801 mAh g^{-1} , respectively, with a fading of 0.31% and 0.20% per cycle after 100 cycles at 4.90 mg cm^{-2} and 4.80 mg cm^{-2} sulfur loading. These degradations are significantly higher in comparison to DMAZF/Sulfur/CNTs cathode. The DMAZF/Sulfur/CNTs electrode is further operated at 7 mg cm^{-2} sulfur loading at 0.5C, leading the smooth cycling performance to 500 cycles with a fading of 0.12%/cycle after 500 cycles. It has been found that the nanopore channels of DMAZF are considerably lower as compared to the Li_2S crystals growth after 118 s observed in in-situ experiments by Hwa et al. [59] Hence, the low fading of DMAZF/Sulfur/CNTs electrode can be strongly attributed to the nano pore channels and Lewis acid-base interactions with PS. The successful strategy of nanoporosity combined with Lewis-acid sites provides multiple advantages for LSBs including: (i) ensures faster Li^+ conduction at a low E/S ratio; (ii) immobilizes the PS by randomly oriented nanopores and suppressing the “shuttle effect”; (iii) enhances the electrochemical kinetics of PS reactions due to the catalytic effect of the Lewis-acid Zn-metal site and (iv) assists in the uniform nucleation of the solid discharge products.

The rate capability of DMAZF/Sulfur/CNTs cathode is evaluated under different C-rates (0.1C–1C) are shown in Fig. 4c and their charge/discharge profiles at the initial cycle of these different C-rates are presented in Fig. 4d. The initial reversible specific capacities at these C-rates are 1006, 801, 752, and 630 mAh g^{-1} , respectively. For the initial cycle, the low Coulombic efficiency is often observed which is attributed to the loss of electrons due to side reactions. As the C-rate increases, the specific capacity decreases and the voltage losses increase. The increase in

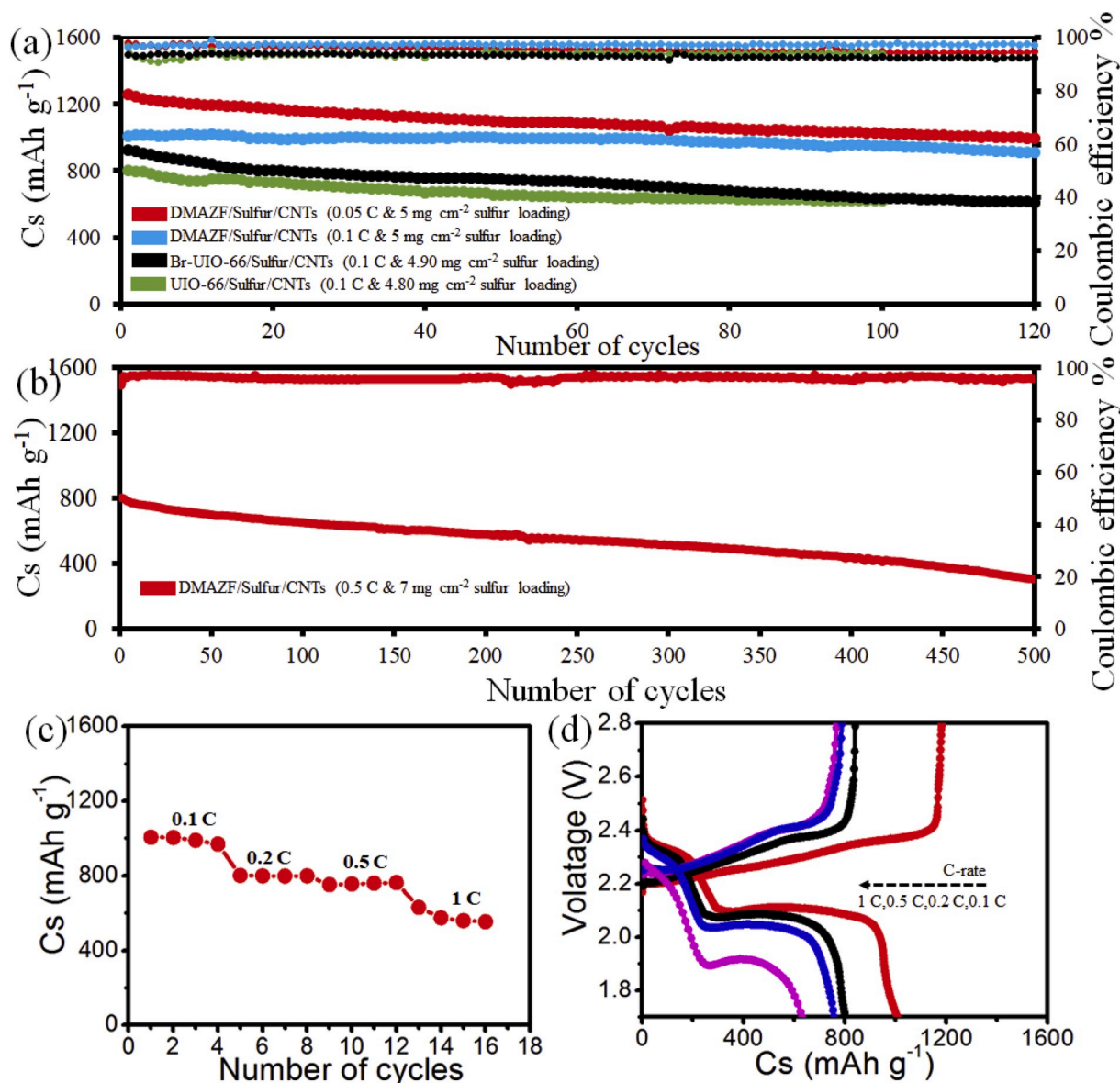


Fig. 4. (a) Cycling performance of DMAZF/Sulfur/CNTs, Br-UIO-66/Sulfur/CNTs, and UIO-66/Sulfur/CNTs. (b) long-run cycling performance of DMAZF/Sulfur/CNTs at high sulfur loading. (c) rate performance of DMAZF/Sulfur/CNTs and its charge/discharge profiles shown in (d).

voltage loss indicates sluggish conversion kinetics over the cycling process. This, in turn, makes it difficult to convert all of the active material – which in turn decreases the specific capacity. The average voltage losses between the charge/discharge curves for the initial cycle of each C-rate (0.1, 0.2, 0.5 and 1C) are 168, 204, 277 and 386 mV respectively. These voltage losses are relatively low in comparison to the existing MOFs studies found in literature [38,39,60]. The high current density for lithiation and delithiation of sulfur electrodes as such becomes challenging as sulfur loading increases. As a result, once current density increases to 1C, the galvanostatic charge/discharge profiles of DMAZF electrode exhibits larger voltage loss.

The surface morphology of the cycled DMAZF and Br-UIO-66 based cathodes are presented in Fig. S6. The Br-UIO-66 based cathode shows structural deformation after the battery cycling. This indicates that the DMAZF can maintain the structural integrity to produce a stable cycling performance. Further, to monitor that cathode is effective in trapping PSs is by observing the Li anode for obvious signs of degradation. To reveal the anode surface changes, the DMAZF, Br-UIO-66, and UIO-66 contained LSBs are pulled apart inside the glovebox and the anodes are investigated by the SEM and EDS analysis. Fig. 5b–d shows the

surface topography of the three different anodes in comparison to the uncycled Li anode (Fig. 5a). The cycled Li anodes (Fig. 5b–d) show a significant surface modification as compared to the uncycled Li anode (Fig. 5a) due to the stripping and plating of Li⁺. More significant morphological changes are observed on the Li anodes of the Br-UIO-66 and UIO-66 containing LSBs. In comparison the Li anode of the DMAZF containing LSB remains comparatively smooth. This further indicates that the DMAZF is more capable of defending the PS migration than the Br-UIO-66 and UIO-66.

This Li anode observation is in good agreement with the PS adsorption tests shown in Fig. S5. A small bottle containing a 0.5 M Li₂S₈ solution was mixed respectively with 40 mg of DMAZF, Br-UIO-66 and UIO-66 and observed for up to three days. After three days the resultant solution of the Br-UIO-66 and UIO-66 sample is still light yellow whereas the resultant solution of the DMAZF sample is clear – indicating a significant PS adsorption ability. Here, the DMAZF immediately absorbs the PS whereas the UIO-66 and Br-UIO-66 take a considerable amount of time. The respective cycling performances (Fig. 4a–b) are also agreed well with the cycled Li anode presented in Fig. 5b–d. The DMAZF serves as an effective protective barrier of PS shuttle due to their nanoporous

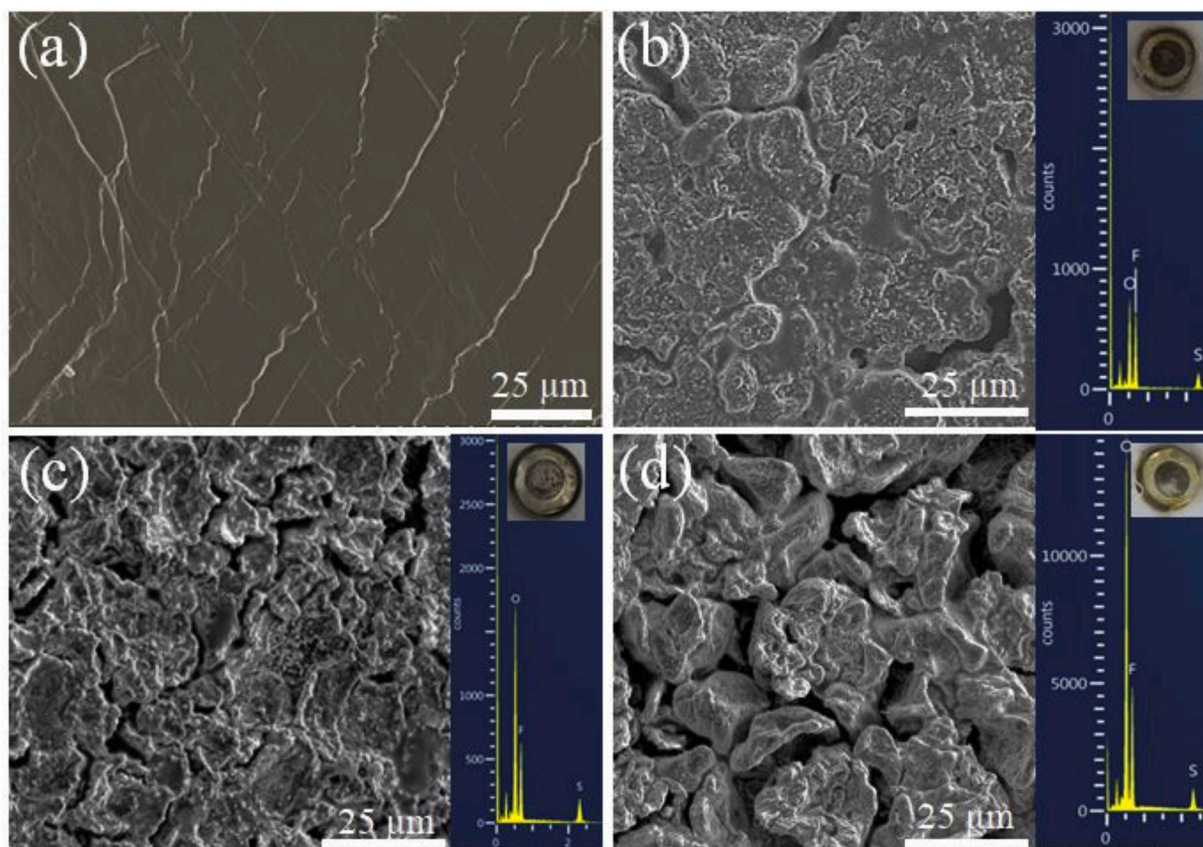


Fig. 5. (a) Uncycled anode. Cycled anode in the presence of (b) DMAZF/CNTs/Sulfur electrode, (c) Br-UIO-66/CNTs/Sulfur electrode and (d) UIO-66/CNTs/Sulfur electrode.

and Lewis acid-base and polar-polar interactive nature leading to high-performance LSBs.

4. Conclusion

In summary, a newly developed nanoporous structure with the electrocatalytic and chemically interactive elements contained DMAZF MOF was first reported for high sulfur content LSB, demonstrating stable electrochemical performance at high sulfur loading. The DMAZF significantly facilitates the Li^+ through its nanopore structures. Furthermore, together with the impact of Lewis-acid interactions, the DMAZF/CNTs/Sulfur electrodes deliver degradation of 0.07% per cycle at 0.1C and 5 mg cm^{-2} sulfur loading. Even with the high sulfur loading of 7 mg cm^{-2} the DMAZF contained LSBs provide an opportunity to achieve 500 cycles, indicating a promising prospect for designing LSBs with a thick electrode for next-generation energy storage devices.

CRediT authorship contribution statement

Masud Rana: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Validation, Visualization, Writing - original draft, Writing - review & editing. **Hydar A. AL-Fayaad:** Formal Analysis, Software, Writing - original draft, Writing - review & editing. **Bin Luo:** Project administration, Resources, Supervision, Validation, Visualization, Writing - original draft, Writing - review & editing. **Tonggen Lin:** Writing - original draft, Writing - review & editing. **Lingbing Ran:** Writing - original draft, Writing - review & editing. **Jack K. Clegg:** Project administration, Resources, Software, Supervision, Validation, Visualization, Writing - original draft, Writing - review & editing. **Ian Gentle:** Project administration, Resources, Software, Supervision, Validation, Visualization, Writing - original draft, Writing - review &

editing. **Ruth Knibbe:** Project administration, Resources, Software, Supervision, Validation, Visualization, Writing - original draft, Writing - review & editing.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.nanoen.2020.105009>.

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